

AI-FDS Student Investigation Brief

Plain-language findings, priority actions, and traceable technical evidence

Case: AI-FDS-20260726-2326-CORRECTED	Generated: 2026-07-27 00:28:24 (Asia/Kuala_Lumpur, GMT+08:00)
Purpose: Educational review of the six evidence records selected in the original Student Brief export.	Evidence reviewed: 6

Investigation Summary

6 Evidence	2 Immediate	2 Review	2 No immediate action
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What should I do next?

1	Pause contact and do not send money, credentials, OTPs, files, or remote access. Stopping the interaction limits further exposure while the evidence is verified.	Immediate
2	Do not disclose the code or password; change exposed credentials and contact the account provider. OTPs and passwords can enable account takeover.	Immediate
3	Verify the person or organization through a separately obtained official channel. An independent channel avoids relying on details supplied by the same contact.	Prompt review

Evidence Review

<i>Evidence 1 - Transcript</i>		No Immediate Action
Result: Legitimate	Evidence score: 20.0%	Time: 2026-07-26 22:15:12
Source: innocent long audio.mp3	Analysis method: Transcript model consensus (0/3 suspicious)	Score meaning: Maximum dashboard concern signal
Why: Transcript suspicious-risk probability: 13.8%. Maximum dashboard concern signal: 20.0%. Key indicators: No explicit indicators saved Relevant excerpt: [Uploaded audio transcript] Next action: Retain the earliest available recording, transcript, timestamps, and unedited working copy.		
<i>Evidence 2 - Transcript</i>		Immediate Action Required
Result: Suspicious	Evidence score: 79.4%	Time: 2026-07-25 22:31:03
Source: [Uploaded audio transcript] Hello, this is the security department fro	Analysis method: Transcript model consensus (3/3 suspicious)	Score meaning: Concern score
Why: Transcript suspicious-risk probability: 79.4%. Stored as risk so dashboard/report history does not treat legitimate confidence as threat. Key indicators: Urgent, Login, Suspended Relevant excerpt: We detected an urgent login attempt on your account today. Next action: Pause contact and do not send money, credentials, OTPs, files, or remote access.		

Evidence 3 - Phone		Review Required
Result: Lower concern	Evidence score: 0.0%	Time: 2026-07-25 14:38:03
Source: Veriphone: success; PenipuMY: success	Analysis method: Phone provider reputation rules	Score meaning: Concern score
Why: Continue with normal caution. Key indicators: Phone ownership unconfirmed Relevant excerpt: +60182874831 Next action: Verify the person or organization through a separately obtained official channel.		

Evidence 4 - Phone		Immediate Action Required
Result: High concern	Evidence score: 72.0%	Time: 2026-07-25 14:35:06
Source: Veriphone: success; PenipuMY: success	Analysis method: Phone provider reputation rules	Score meaning: Concern score
Why: Pause the interaction and verify independently. Key indicators: 5 police report(s) found, Fraud flag returned, Spam flag returned Relevant excerpt: +60123456789 Next action: Pause contact and do not send money, credentials, OTPs, files, or remote access.		

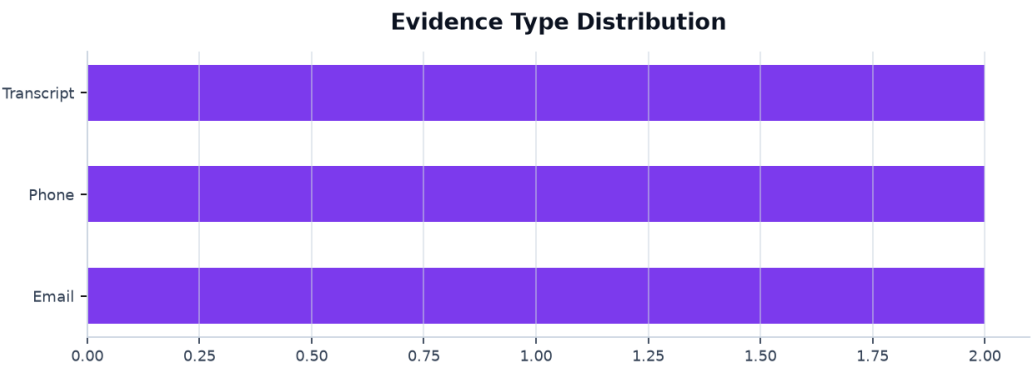
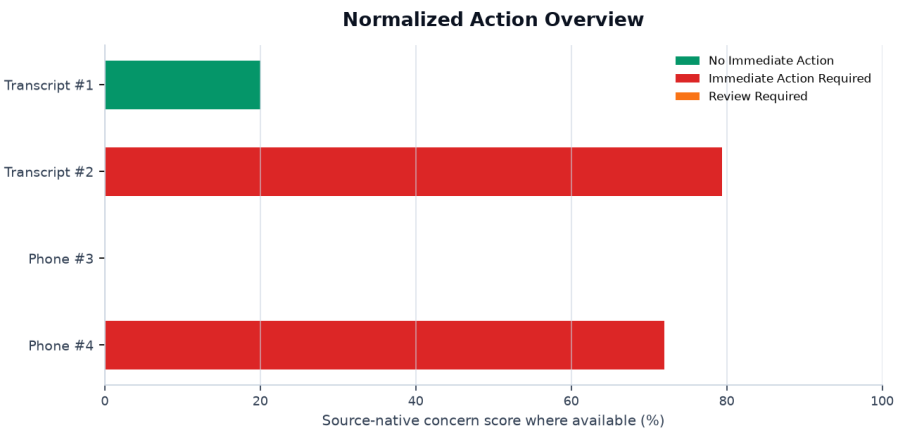
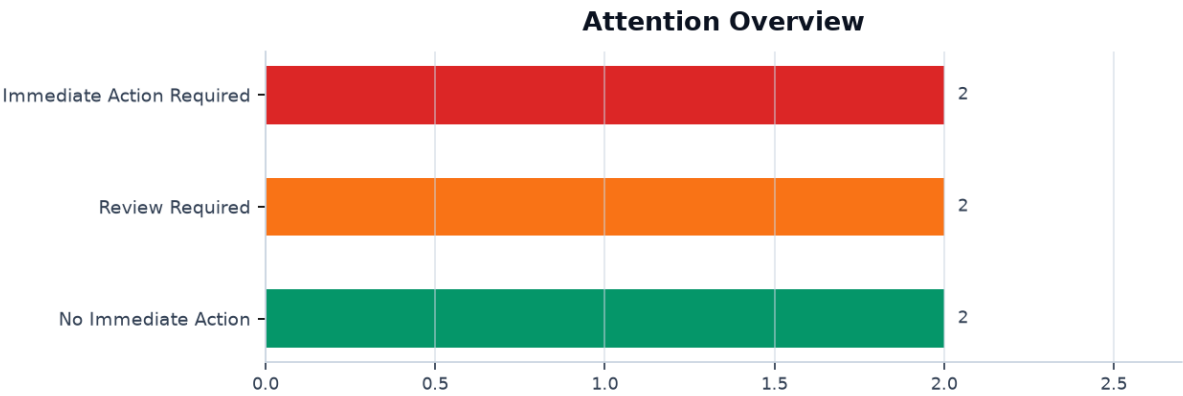
Evidence 5 - Email		Review Required
Result: Suspicious	Evidence score: N/A	Time: 2026-07-18 22:30:28
Source: Subject: URGENT: Verification Required - Immediate Security Action Req	Analysis method: Email Random Forest	Score meaning: Concern score
Why: The evidence is incomplete, conflicting, or suspicious enough to verify. Key indicators: No explicit indicators saved Relevant excerpt: 1. Next action: Verify the person or organization through a separately obtained official channel.		

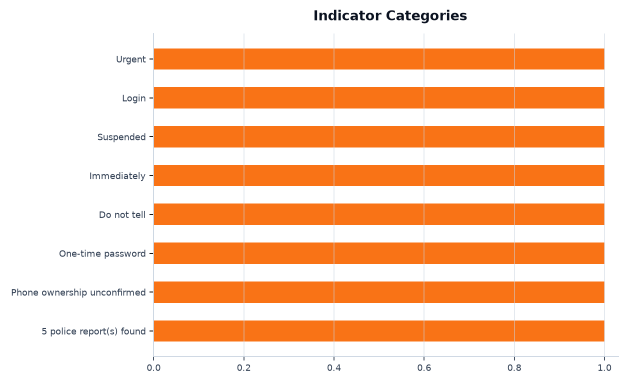
Evidence 6 - Email		No Immediate Action
Result: Legitimate	Evidence score: N/A	Time: 2026-07-18 22:29:40
Source: Subject: Project Kickoff & Sync Meeting - Q3 Roadmap Hi Team, I hope	Analysis method: Email Support Vector Machine	Score meaning: Concern score
Why: The available evidence does not indicate immediate action, but normal caution still applies. Key indicators: No explicit indicators saved Relevant excerpt: Hi Team, Next action: Retain the original message, complete headers, attachments, URLs, and timestamps.		

Combined Investigation Findings

Finding	Evidence	Priority
5 police report(s) found	Phone #4	High
5 total report indicator(s)	Phone #4	High
Do not tell	Transcript #2	High
Fraud flag returned	Phone #4	High
Immediately	Transcript #2	High

Visual Evidence Summary





Interpretation note: email probabilities, transcript/audio risk signals, and phone evidence scores use different methods and are not directly comparable.

Reviewer Note

Verify suspicious contact through official channels, do not share OTP/passwords, preserve the evidence, and report urgent financial or identity-related requests to the relevant campus or organisation support team.

Technical Appendix

Investigation-specific output is grouped under its evidence record. Shared training evaluation metrics appear once in a separate reference section.

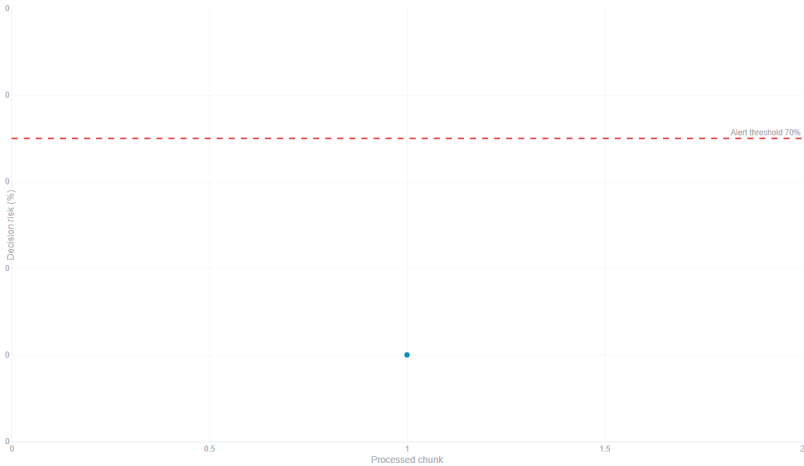
Evidence-Specific Supporting Visuals and Data

Evidence 1 - Transcript		No Immediate Action
Result: Legitimate	Source: innocent long audio.mp3	Time: 2026-07-26 22:15:12

Supporting Visuals and Data

1. [Chart] Recording 1 - Decision Risk Timeline

Decision risk for every processed chunk in this recording.



clip	chunk	decision	risk
1	1	Lower threat - limited voice evidence	20.0

2. [Table] Recording 1 - Chunk Investigation Table

Clip	Chunk	Time	Decision	Decision Risk
1	1	22:15:11	Lower threat - limited voice evidence	20.0%

Raw Blend	Voice Evidence	Raw Voice AI	Transcript	Behavioral Evidence
59.6%	18.3%	99.9%	15.6%	54.5%

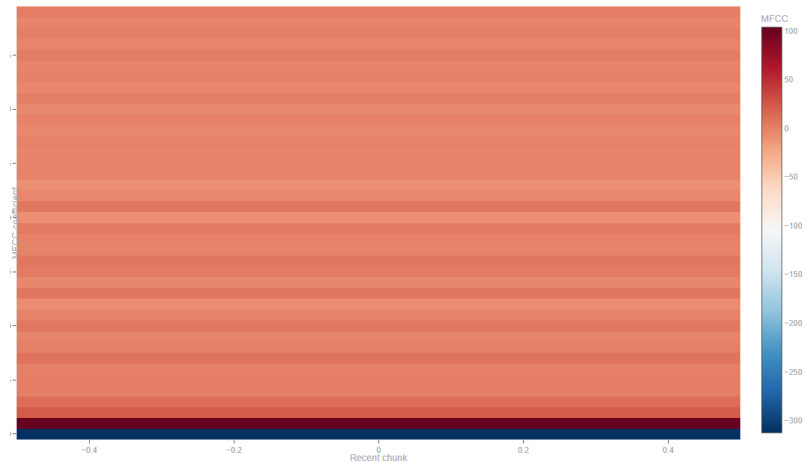
Raw Behavioral RF	Voice Reliability	Behavioral Reliability	Content Reliability	Content Level
63.3%	87.5%	86.0%	57.5%	Lower scam-content concern

Voice Level	Speech quality	Whisper	Language check	Detected text
Weak voice-authenticity evidence	Usable speech-like audio	Transcript accepted	en	Hello, this is me trying to tell you about your call status warranty. Please message me back whenever you can. You have something like that.

Flags
-

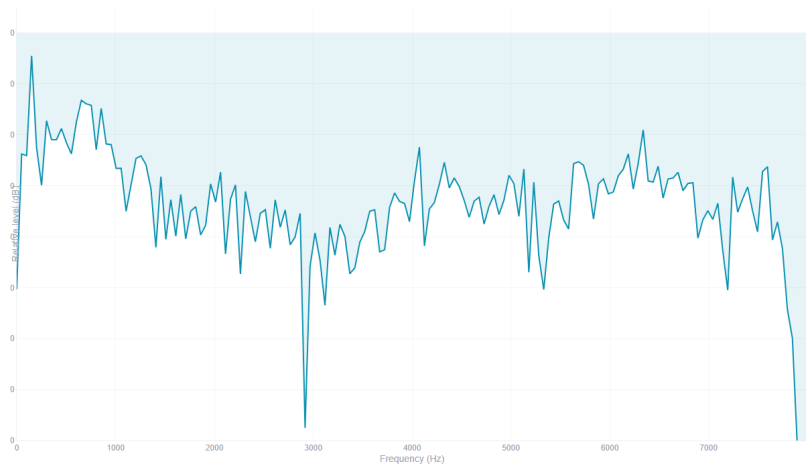
3. [Chart] Recording 1 - MFCC Feature Heatmap

Each heatmap column is a recent chunk, and each row is an MFCC coefficient: a compact fingerprint of vocal tone, timbre, and spectral shape used by the voice model. The heatmap supports the raw voice score, but reliability checks decide whether that pattern is strong enough to trust. Current reading: 1 chunk(s), 11.7s latest audio, weak voice-authenticity evidence. Reliability limits: none.



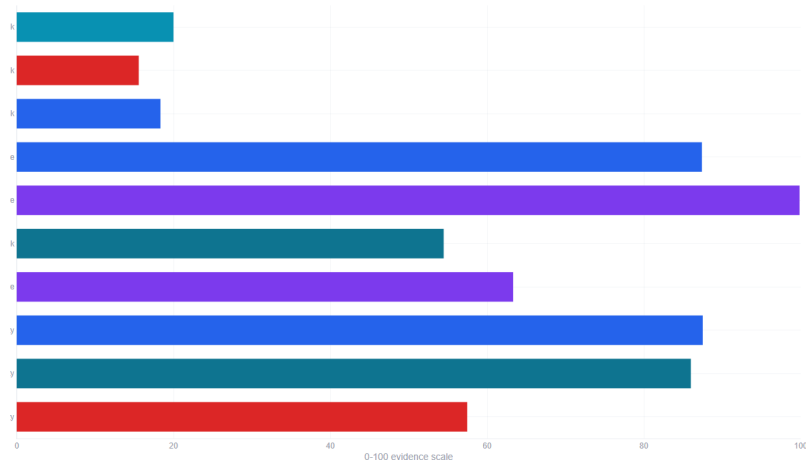
4. [Chart] Recording 1 - Frequency Spectrum

Relative frequency levels from the latest processed chunk.



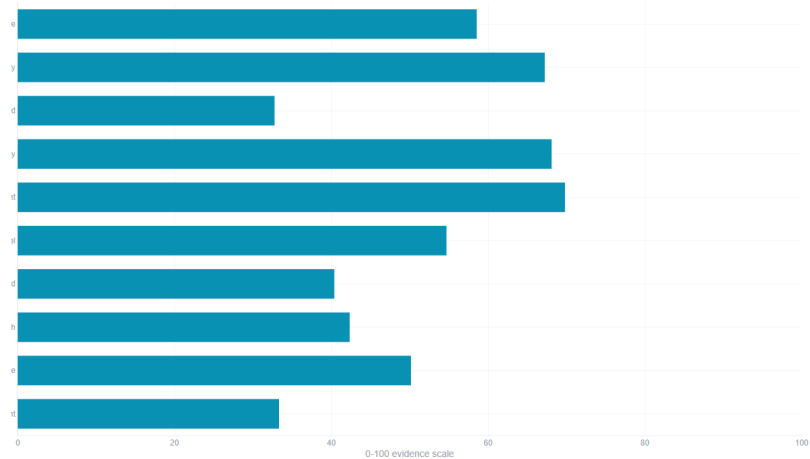
5. [Chart] Recording 1 - Voice Evidence Metrics

This chart separates raw model probabilities from reliability-weighted evidence. Primary voice evidence source: Trained voice evidence calibrator. Conclusion: Lower threat - limited voice evidence; the raw voice model is high, but it is not strong evidence after reliability checks; behavioral RF moved from 63.3% raw to 54.5% evidence.



6. [Chart] Recording 1 - Behavioral RF Signal Metrics

Behavioral RF looks at duration, voiced activity, silence, energy, spectral shape, pauses, and speech rate. The bars are normalized so mixed units can be compared visually; hover details keep the raw measurement. Conclusion: RF is 63.3% raw, 54.5% after 86.0% reliability. Reliability limits: none.



7. [Table] Recording 1 - Behavioral Feature Values

Behavior Feature	Value
Duration Seconds	11.7120
Rms Energy Mean	0.0341
Rms Energy Std	0.0279
Zero Crossing Rate Mean	0.1094
Zero Crossing Rate Std	0.1174
Spectral Centroid Mean	1615.6310
Spectral Centroid Std	1060.3790
Spectral Bandwidth Mean	1694.0490
Spectral Rolloff Mean	3151.6106
Silence Ratio	0.3276
Speech Activity Ratio	0.6724
Pause Count	4.0000
Estimated Speech Rate	4.0130

8. [Chart] Transcript Live Model Comparison

Suspicious-risk probability for every selected transcript model.



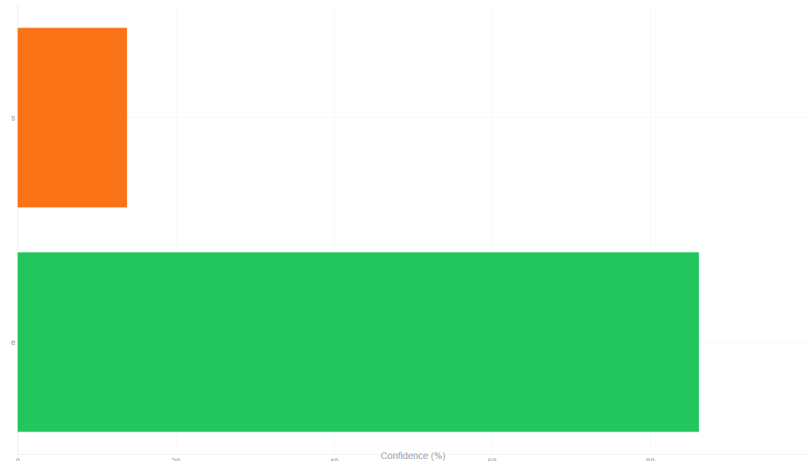
Model	Model Key	Metrics Model	Prediction	Risk Score
DistilBERT	distilbert	DistilBERT	Legitimate	12.56
SVM	svm	SVM	Legitimate	12.6
Naive Bayes	nb	Naive Bayes	Legitimate	16.34
Confidence	Accuracy	Precision	Recall	F1 Score
87.44	83.62	86.36	74.51	80.0
87.4	81.9	81.25	76.47	78.79
83.66	80.17	85.0	66.67	74.73
ROC-AUC	Training Time (s)	Prediction Time (ms)	Engine	
91.28	590.212	322.4884	Transcript DistilBERT	
90.86	0.078	0.0382	Transcript SVM	
91.19	0.004	0.0039	Transcript Naive Bayes	

9. [Table] Transcript Live Model Comparison Table

Model	Model Key	Metrics Model	Prediction	Risk Score
DistilBERT	distilbert	DistilBERT	Legitimate	12.56
SVM	svm	SVM	Legitimate	12.6
Naive Bayes	nb	Naive Bayes	Legitimate	16.34
Confidence	Accuracy	Precision	Recall	F1 Score
87.44	83.62	86.36	74.51	80.0
87.4	81.9	81.25	76.47	78.79
83.66	80.17	85.0	66.67	74.73
ROC-AUC	Training Time (s)	Prediction Time (ms)	Engine	
91.28	590.212	322.4884	Transcript DistilBERT	
90.86	0.078	0.0382	Transcript SVM	
91.19	0.004	0.0039	Transcript Naive Bayes	

10. [Chart] Transcript Consensus Probability

Legitimate and suspicious probabilities represented by the live dashboard.



Field	Value
Legitimate	0.8616666666666667

Field	Value
Suspicious	0.13833333333333334

11. [Table] Transcript Explainability Evidence

Indicator	Type	Model Weight	Impact	Intention
tell	SVM: Legitimate wording	-0.2094	Medium	Shows which words pushed a transparent baseline model up or down; supporting evidence only.
status	SVM: Legitimate wording	-0.1719	Medium	Shows which words pushed a transparent baseline model up or down; supporting evidence only.
message	SVM: Legitimate wording	-0.1701	Low	Shows which words pushed a transparent baseline model up or down; supporting evidence only.
like	SVM: Legitimate wording	-0.0812	Low	Shows which words pushed a transparent baseline model up or down; supporting evidence only.
trying	SVM: Suspicious wording	0.0438	Low	Shows which words pushed a transparent baseline model up or down; supporting evidence only.
hello	SVM: Legitimate wording	-0.0022	Low	Shows which words pushed a transparent baseline model up or down; supporting evidence only.
tell	Naive Bayes: Legitimate wording	-0.3695	High	Shows which words pushed a transparent baseline model up or down; supporting evidence only.
message	Naive Bayes: Legitimate wording	-0.344	High	Shows which words pushed a transparent baseline model up or down; supporting evidence only.
hello	Naive Bayes: Legitimate wording	-0.2759	High	Shows which words pushed a transparent baseline model up or down; supporting evidence only.
status	Naive Bayes: Legitimate wording	-0.2507	Medium	Shows which words pushed a transparent baseline model up or down; supporting evidence only.
like	Naive Bayes: Legitimate wording	-0.1475	Low	Shows which words pushed a transparent baseline model up or down; supporting evidence only.
trying	Naive Bayes: Legitimate wording	-0.0196	Low	Shows which words pushed a transparent baseline model up or down; supporting evidence only.

12. [Text] Original Combined Transcript

Exact labelled transcript input submitted to the text models.

[Uploaded audio transcript] Hello, this is me trying to tell you about your call status warranty. Please message me back whenever you can. You have something like that.

Evidence 2 - Transcript		Immediate Action Required
Result: Suspicious	Source: [Uploaded audio transcript] Hello, this is the security department fro	Time: 2026-07-25 22:31:03

Supporting Visuals and Data

No investigation-specific dashboard artifacts were captured for this evidence record.

Evidence 3 - Phone		Review Required
Result: Lower concern	Source: Veriphone: success; PenipuMY: success	Time: 2026-07-25 14:38:03

Supporting Visuals and Data

No investigation-specific dashboard artifacts were captured for this evidence record.

Evidence 4 - Phone		Immediate Action Required
Result: High concern	Source: Veriphone: success; PenipuMY: success	Time: 2026-07-25 14:35:06

Supporting Visuals and Data

No investigation-specific dashboard artifacts were captured for this evidence record.

Evidence 5 - Email		Review Required
Result: Suspicious	Source: Subject: URGENT: Verification Required - Immediate Security Action Req	Time: 2026-07-18 22:30:28

Supporting Visuals and Data

No investigation-specific dashboard artifacts were captured for this evidence record.

Evidence 6 - Email		No Immediate Action
Result: Legitimate	Source: Subject: Project Kickoff & Sync Meeting - Q3 Roadmap Hi Team, I hope	Time: 2026-07-18 22:29:40

Supporting Visuals and Data

No investigation-specific dashboard artifacts were captured for this evidence record.

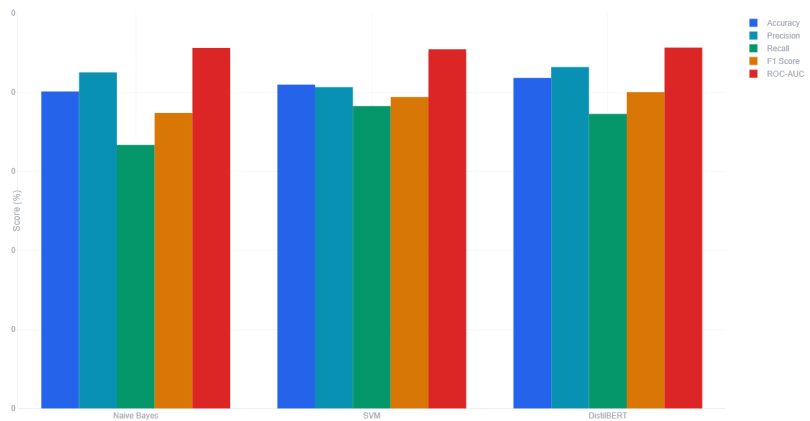
Shared Model Reference Metrics

These are model evaluation references, not measurements produced by a single investigation.

1. [Chart] Performance Metrics

Used by: Evidence 1 - Transcript

Evaluation tab 1 of 3: saved transcript training metrics.



Model	Accuracy	Precision	Recall	F1 Score
Naive Bayes	80.17	85.0	66.67	74.73
SVM	81.9	81.25	76.47	78.79
DistilBERT	83.62	86.36	74.51	80.0

ROC-AUC	Training Time (s)	Prediction Time (ms)	True Positive	False Positive
91.19	0.004	0.0039	34	6
90.86	0.078	0.0382	39	9
91.28	590.212	322.4884	38	6

True Negative	False Negative
59	17
56	12
59	13

2. [Table] Performance Metrics data

Used by: Evidence 1 - Transcript

Model	Accuracy	Precision	Recall	F1 Score
Naive Bayes	80.17	85.0	66.67	74.73
SVM	81.9	81.25	76.47	78.79
DistilBERT	83.62	86.36	74.51	80.0

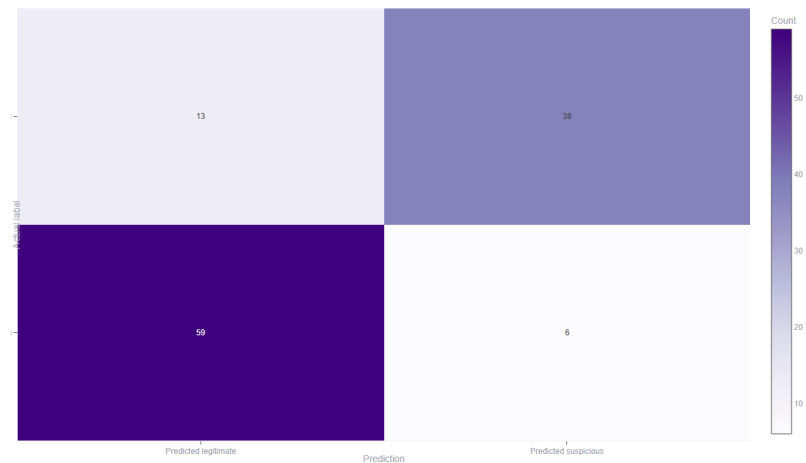
ROC-AUC	Training Time (s)	Prediction Time (ms)	True Positive	False Positive
91.19	0.004	0.0039	34	6
90.86	0.078	0.0382	39	9
91.28	590.212	322.4884	38	6

True Negative	False Negative
59	17
56	12
59	13

3. [Chart] Confusion Matrix Heatmap

Used by: Evidence 1 - Transcript

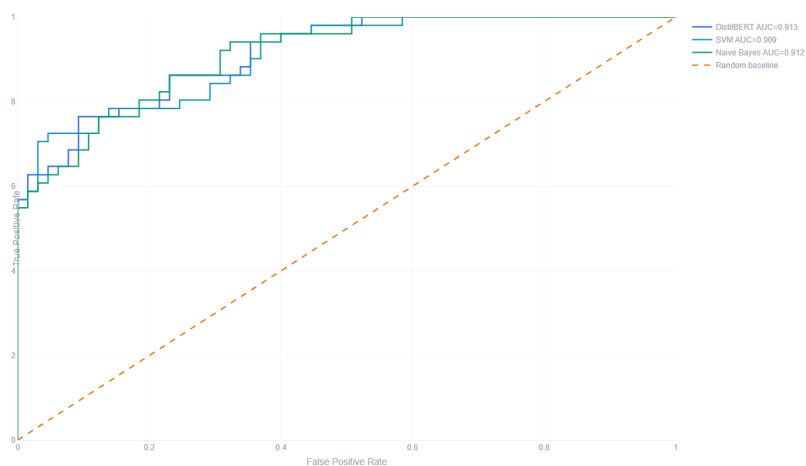
Evaluation tab 2 of 3 for the recommended transcript model.



4. [Chart] ROC-AUC Curve

Used by: Evidence 1 - Transcript

Evaluation tab 3 of 3 for selected transcript models.



Evidence Register

#	Type	Time	Source	Input SHA-256	AI analysis
1	Transcript	2026-07-26 22:15:12	innocent long audio.mp3	fc272fb45527e47dad52e9e54a36becb048f06b3a b1b9c1caf2531a0aa3353e8	Transcript model consensus (0/3 suspicious)
2	Transcript	2026-07-25 22:31:03	[Uploaded audio transcript] Hello, this is the security department fro	Legacy - not captured	Transcript model consensus (3/3 suspicious)
3	Phone	2026-07-25 14:38:03	Veriphone: success; PenipuMY: success	Legacy - not captured	Phone provider reputation rules
4	Phone	2026-07-25 14:35:06	Veriphone: success; PenipuMY: success	Legacy - not captured	Phone provider reputation rules
5	Email	2026-07-18 22:30:28	Subject: URGENT: Verification Required - Immediate Security Action Req	Legacy - not captured	Email Random Forest
6	Email	2026-07-18 22:29:40	Subject: Project Kickoff & Sync Meeting - Q3 Roadmap Hi Team, I hope	Legacy - not captured	Email Support Vector Machine

Legacy limitation: 5 selected record(s) predate immutable evidence snapshots.

Scope and References

Educational scam-awareness support only. Results support review and do not prove identity, intent, legal liability, or authenticity.

EC-Council CEH - Social Engineering, Phishing, AI Impersonation, and Countermeasures

<https://www.eccouncil.org/train-certify/certified-ethical-hacker-ceh/>

NIST SP 800-61 Rev. 3 - Incident Response Recommendations and Considerations

<https://csrc.nist.gov/pubs/sp/800/61/r3/final>

SWGDE Requirements for Report Writing in Digital and Multimedia Forensics

<https://www.swgde.org/documents/published-complete-listing/18-q-002-swgde-requirements-for-report-writing-in-digital-and-multimedia-forensics/>

SWGDE Best Practices for Digital Evidence Collection

<https://www.swgde.org/documents/published-complete-listing/18-f-002-best-practices-for-digital-evidence-collection/>

NISTIR 8312 - Four Principles of Explainable Artificial Intelligence

<https://www.nist.gov/publications/four-principles-explainable-artificial-intelligence>

CISA Secure Our World - Recognize and Report Phishing

<https://www.cisa.gov/secure-our-world>

FTC Consumer Advice - AI Voice Cloning and Family Emergency Scams

<https://consumer.ftc.gov/articles/scammers-use-fake-emergencies-steal-your-money>

FCC Consumer Complaints - Unwanted Calls, Texts, and Caller ID Spoofing

<https://consumercomplaints.fcc.gov/hc/en-us/articles/115002234203-Unwanted-Calls-Texts-Phone>